

5	(a)	$R = \rho l / A$	C1	
		$= (5.1 \times 10^{-7} \times 0.50) / \pi(0.18 \times 10^{-3})^2 = 2.5 \text{ (2.51)} \Omega$	M1	[2]
	(b)	(i)	resistance of CD = $8 \times$ resistance of AB = $20 \text{ (}\Omega\text{)}$	C1
			circuit resistance = $[1/5.0 + 1/20]^{-1} = 4.0 \text{ (}\Omega\text{)}$	C1
			current = $V/R = 6.0/4.0$	C1
			$= 1.5 \text{ A}$	A1 [4]
		(ii)	power in AB = $I^2 R$	or power = V^2/R C1
			$= (1.2)^2 \times 2.5 = 3.6 \text{ W}$	$= (3.0)^2/2.5 = 3.6 \text{ W}$ A1 [2]

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(iii) potential drop A to M = $1.25 \times 1.2 = 1.5 \text{ V}$

M1

potential drop C to N = 3.0 V

p.d. MN = 1.5 V

A1

[2]